



POLITECNICO
MILANO 1863

Hypermedia Applicaton Design Report

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April 2025

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1 Abstract

The purpose of this design report is to provide a comprehensive overview of the key elements and decisions made during the website implementation for a yoga center. The report meticulously outlines the primary design choices made throughout the project, supported by relevant models, schemas, and use case scenarios. Furthermore, it presents the conceptual and logical structure of the database, which underpins user interaction, class scheduling, subscription management, and content delivery functionalities.

2 Conceptual design

This section presents an exhaustive summary of the conceptual design principles applied to the website. It meticulously examines the decision-making process governing the content selection, its structuring into pages, and the seamless navigation between them. The subsequent sections delve into the intricacies of these decisions, providing a comprehensive explanation and analysis.

2.1 C-IDM Diagram

In this subsection, the Content Interaction Dialogue Model (C-IDM) is introduced as a model that represents the website as a conversation between the user and the application. This model elucidates the nature of the dialogue, including its classification, inter-relationships, and grouping of conversation topics. Additionally, it provides insights into the content that can be discussed within each category.

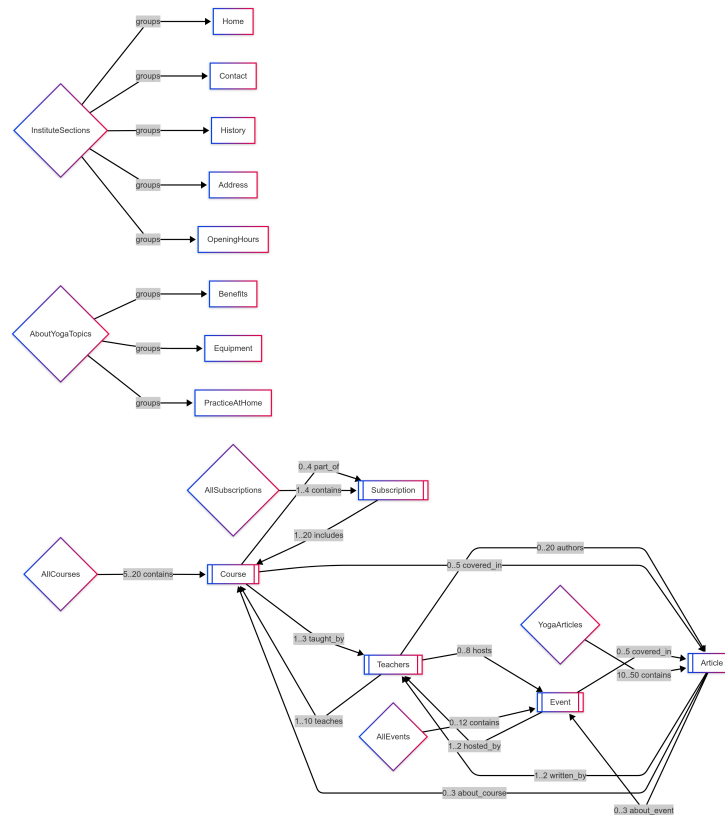


Figure 1: C-IDM Diagram

2.2 Content-in-the small Tables

2.2.1 Topics

Topic:	Home
Title:	text (max 50 chars)
NewCustomerOffer:	text
FeaturedPanel:	text
DiscoverCourseLink:	URL
ContactLink:	URL

Table: Home

Topic:	Contact
ContactInfo:	text (max 200 chars)
ContactFormFiels:	List of text
SocialMediaLinks:	List of text

Table: Contact

Topic:	Histroy
Title:	text (max 200 chars)
Content:	text (max 1000 chars)
FounderImages:	List of Image

Table: History

Topic:	Address
AddressText:	text (max 200 chars)
MapEmbed:	interactive map
PublicTransportInfo:	text (max 500 chars)
OpenHoursLink:	URL

Table: History

Topic:	PracticeAtHome
Title:	text (max 200 chars)
Content:	text (max 500 chars)
Exercice:	text (max 500 chars)
OpenHoursLink:	URL
VideoLinks:	List of URLs
SafetyGuidelines:	text (max 500 chars)

Table: Practice at home

Topic:	OpeningHours
WeeklySchedule:	List of (Day, CourseName, Time) (max 200 chars)
PrivateCourseBookingLink:	URL

Table: Opening Hours

Topic:	Benefit
Title:	text (max 200 chars)
Content:	text (max 500 chars)
FeatureList:	List of (Benefit, Description) (max 500 chars)
RelatedLink:	URL

Table: Benefit

Topic:	Equipement
Title:	text (max 200 chars)
Content:	text (max 500 chars)
FeatureList:	List of (Benefit, Description) (max 500 chars)
RelatedLink:	URL

Table: Equipement

2.2.2 Kind of Topic

Kind of topic:	Course
Title:	text (max 200 chars)
Image:	image
Description:	text (max 500 chars)
Goal:	text (max 200 chars)
Timetable:	List of (Day, Time)
Price:	currency
IntroVideoUrl:	URL
TaughtBy:	List of Teachers, Link
PartOfSubscription:	List of (Subscription, Link)
CoveredInArticle:	List of (Article, Link)

Table: Course

Kind of Topic:	Subscription
Name:	text (max 200 chars)
Type:	text (max 200 chars)
TotalCourse:	integer
Price:	Currency
Link:	URL
ContactLink:	URL
Details:	text (max 500 chars)
CoursesIncluded:	List of (CourseTitle, Link)

Table: Equipement

Kind of Topic:	Event
Title:	text (max 200 chars)
Image:	Image
Date: Timestamp	
Description:	text (max 500 chars)
Location:	text (max 500 chars)
BookingLink:	URL
HostedBy:	List of (Teachers, Link)
CoveredInArticles: List of (Article, Link)	

Table: Event

Kind of Topic:	Teacher
Name:	text (max 200 chars)
Photo:	Image
Biography:	text (max 500 chars)
Certificates:	List of Image or Text
CoursesTaught:	List of (Course, Link)
SocialMedias:	List of URL
HostedEvents:	List of (Event, Link)
List of (Article, Link)	AuthoredArticles:

Table: Teacher

Kind of Topic:	Article
Title:	text (max 200 chars)
ThumbnailImage:	Image
Snippet:	text (max 200 chars)
FullText:	text
Authors:	List of Text or (Teachers, Link)
Date (Timestamp)	PublishDate:
AboutCourses:	List of Course, Link
AboutEvents:	List of (Event, Link)

Table: Article

2.2.3 Group of Topics

Group:	AllCourses
Title:	text (max 200 chars)
Description:	text (max 500 chars)
MembersPreview:	List of (Course, Thumbnail, Title, Price)

Table: All Courses

Group:	AllSubscription
Title:	text (max 200 chars)
Description:	text (max 500 chars)
MembersPreview:	List of (Subscription, Name, TotalCourses, Price)

Table: All Subscriptions

Group:	AllEvents
Title:	text (max 200 chars)
Description:	text (max 500 chars)
MembersPreview:	List of (Event, Thumbnail, Title, Date)

Table: All Events

Group:	AllTeachers
Title:	text (max 200 chars)
Description:	text (max 500 chars)
MembersPreview:	List of (Teacher, Names, CoursesTaught)

Table: All Teachers

Group:	InstitueSection
Title:	text (max 200 chars)
Description:	text (max 500 chars)
MembersPreview:	List of (SectionTitle, Snippet)

Group:	AboutYogaTopics
Title:	text (max 200 chars)
Description:	text (max 500 chars)
MembersPreview:	List of (TopicTitle, Link)

Table: About Yoga Topics

Group:	AllYogaArticles
Title:	text (max 200 chars)
Description:	text (max 500 chars)
MembersPreview:	List of (Article, Thumbnail, Title, PublishDate)

Table: All Yoga Articles

3 Final Commented Screenshots

tabularx caption

4 Interaction scenarios

This section presents concrete examples demonstrating how users will interact with the web application. It meticulously outlines the sequence of actions necessary to access the site's content and features. Each interaction scenario embodies a realistic use case and is accompanied by a descriptive narrative and visual aids, with interactive elements explicitly highlighted.

4.1 Use case 1

4.1.1 Textual narrative

Title:	Find how to reach the center in metroline
Profile:	A potential customer seeking to access the yoga center via the metro system
Goal:	Their goal is to find the metro route from their current location to the yoga center
Context:	They access the web application from any personal device
Tasks:	• They access the <i>Home page</i> of the website • They navigate to the <i>Institute</i> page using the navigation bar. Then, they click on the <i>Where to find us?</i> button, which leads them to the <i>Location</i> page, where an interactive map displays the center's address. Alternatively, they can access the <i>Location</i> page directly via the navigation bar.

Use case: Access the center by using metroline

4.1.2 Interaction Flow

4.2 Use case 2

4.2.1 Textual narrative

Title:	Find somewhere to contact the center
Profile:	A potential customer seeking to reach the yoga center
Goal:	Their goal is to find a way to contact the center
Context:	They access the web application from any personal device
Tasks:	• They access the <i>Home page</i> of the website • They navigate to the <i>Contact</i> page using the navigation bar. • They have the capability to compose messages via the contact form.

Use case: Reach the center

4.2.2 Interaction Flow

4.3 Use case 3

4.3.1 Textual narrative

Title:	You want to know some exercises you can do at home
Profile:	A user seeking to practice at home
Goal:	Their goal is to find exercises and instructions to practice yoga at home
Context:	They access the web application from any personal device
Tasks:	• They access the <i>Home page</i> of the website • They navigate to the <i>About</i> page using the navigation bar. Then, they click on the <i>Start now</i> button, which leads them to the <i>Practice at home</i> page, where they can find a list of exercises, tutorials videos and some guidelines. Alternatively, they can access the <i>Practice at home</i> page directly via the navigation bar.

Use case: Practice yoga at home

You are looking for something to do on Tuesday afternoon, find which courses are given You are looking for an event on next Hollydays, find which event takes place between June 7th and June 12th Your friend recommended you their favorite teacher, Ms. B, and you want to try some courses with her. Find her courses and then find the conditions to participate at 3 courses of her at the best price.

5 Database Design

This section presents the design of the database for the website to be implemented. Indeed, it manages students, teachers, courses, subscriptions, feedback, equipment, events, and articles. The schema includes detailed relationships between entities, such as students enrolling in courses, courses requiring specific equipment, and subscriptions granting access to various learning materials. JSON fields are used for flexible data such as certificates, goals, and social media links.

5.1 Explanation of Relationship Symbols

The following notation is used in the Entity-Relationship (ER) diagrams, following the Mermaid syntax conventions that you can check on : mermaid.js.org

5.2 Entity-Relation Diagram

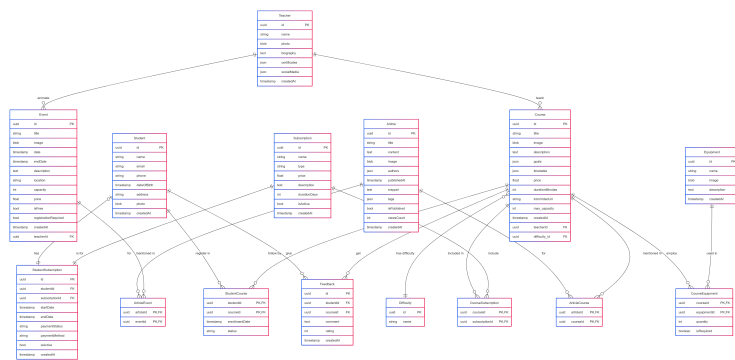


Figure 2: E-R Diagram